



Missing flight feathers

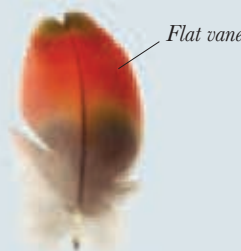
Feather types



Soft filaments

Down feather

With their fluffy filaments, down feathers trap a layer of warm air next to a bird's skin and help to cushion and protect its body.



Flat vane

Contour feather

These feathers have soft filaments at their base and flat tips that overlap to give the bird's body a streamlined surface.



Stiff quill

Flight feather

The smooth vane of a flight feather is very light with a central, rigid spine, or quill, bearing hundreds of tiny strands, called barbs.

Running repairs

A bird's delicate plumage becomes tattered with use—the wing feathers, in particular, get a tremendous amount of wear and tear. To stay in top condition, birds shed, or molt, their feathers regularly and grow new ones in their place. Like most birds, this wedge-tailed eagle molts its flight feathers in a set pattern, a few at a time. Water birds, however, often lose their flight feathers all at once. They cannot fly until their new feathers have grown back, so the birds molt in remote places, staying a safe distance from predators on the shore.

All zipped up

Viewed up close, this Eurasian jay's flight feather looks like a comb with tiny teeth, or barbs, set in neat rows that almost touch. The barbs are armed with smaller branches, called barbules, which lock together with flexible hooks. The barbs and interlocking barbules make up the feather's flat surface, known as the vane, and enable the feather to bend and stretch as the wing beats up and down in flight.

Microscopic view of interlocking barbules



Colorful display

Birds include some of the most brilliantly colored animals in the natural world. In many species, such as these birds of paradise, the males use their striking plumage as a signal to attract a mate. Most feather colors are produced by chemical pigments, which are stored inside a bird's feathers as they grow. Shiny, or iridescent, colors—like those of the hummingbird (see p. 40)—are created in a different way, when light reflects off the microscopic structures on the surface of feathers.